

8

Predictions of Future Climate Change

In Chapter 6, we discussed the concept of radiative forcing, which is an imposed change in planetary energy balance. In response, the planet's temperature adjusts so as to restore energy balance, with the climate sensitivity (Section 6.3) determining how much warming is required. Thus, predictions of future climate require predictions of how radiative forcing will evolve in the future combined with an estimate of the Earth's climate sensitivity.

Predicting future radiative forcing basically comes down to predicting how much greenhouse gas and aerosol will be emitted into the atmosphere each year from human activities. Such projections, known as *emissions scenarios*, therefore form the backbone of our predictions of climate change. In this chapter, I describe how they are constructed and what they tell us about our future climate.

8.1 The Factors That Control Emissions

At its simplest, the amount of greenhouse gas released by a society is determined by the total amount of goods and services consumed by that society. This is true because the production of any good or service – be it a car, an iPhone, a university lecture, a cheeseburger, or an hour of tax consulting – requires energy. And energy today is mostly derived from the combustion of fossil fuels, which leads to the release of carbon dioxide to our atmosphere. The emissions of other greenhouse gases and aerosols also generally scale with the amount of consumption, although the causal linkages may not be as direct.

The total value of goods and services produced by an economy is known as the *gross domestic product*, abbreviated GDP. Thus, total emissions by a society are basically set by that society's GDP. If the GDP doubles, then we expect emissions to roughly double. This strong link between GDP and emissions can be seen during recessions. During the economic downturn of the late 2000s, global carbon emissions posted their biggest drop in more than 40 years as the global recession froze economic activity and slashed energy use around the world. As I write this in late 2020, during the coronavirus pandemic, we are experiencing another drop in emissions due to lower consumption of goods and services caused by people staying home.

Rather than consider GDP as a whole, it is useful to break it into the product of two factors: population and affluence. It should be obvious that GDP scales with population. Every person in a society consumes goods and services, so if the population doubles (and everything else remains the same), then total GDP will also double. Emissions will therefore be proportional to population.

In addition to the number of people, how rich each person is also matters because, as people get richer, they consume more. To illustrate the effect of affluence on GDP and emissions, consider three families. The first is a family of four who live as subsistence farmers in sub-Saharan Africa. This family lives in a small one-room house without electricity or running water. They do not own a car and are too poor to buy anything but the bare necessities of life. They farm by hand or with a draft animal. Because the members of this family are so poor and consume so little, they are responsible for little greenhouse-gas emissions.

Now consider a family of four near the bottom of the economic ladder in the United States. They live in an apartment and they own one car. Their apartment is not air-conditioned; they own a television and one or two heavy-duty electrical appliances, such as an oven. Compared with the subsistence farming family in Africa, this family is far richer and consumes far more, and is therefore responsible for more greenhouse-gas emissions.

But their emissions are far below an upper-class family of four in the United States. This family lives in a 4,000 square foot single-family house and owns three cars (for the husband, wife, and a teenage child). The house is air-conditioned and has televisions in almost every room, several computers, VCRs, game consoles, and a rich assortment of electrical appliances. The family flies to several vacation destinations every year. Because of the significant consumption allowed by their affluence, this family is responsible for more emissions than the poorer US family and many, many times the emissions of the subsistence farming family.

This wealth effect leads to enormous disparities in emissions per person. In the United States, emissions were 16.2 tonnes of carbon dioxide per person in 2017.¹ Emissions in China are 6.9 tonnes per person – about 45 percent of US per capita emissions. However, China's population is so large that they nevertheless lead the world in total carbon emissions. Emissions in Nigeria are 0.6 tonnes per person – about one-thirtieth of the United States – reflecting that country's poverty.

Aside 8.1: Units of carbon dioxide emissions

In our discussions of the carbon cycle in Chapter 5, we used gigatonnes of carbon (GtC) as our unit. As we discussed at the time, this unit counts just the mass of the carbon in the carbon dioxide molecule. This is done because the carbon atom is incorporated in different molecules in the different reservoirs: it might be in a carbohydrate molecule when stored in the land biosphere or in a carbonate when stored in the ocean, or a carbon dioxide molecule when stored in the atmosphere. It therefore makes the most sense to consider just the mass of carbon as carbon flows around the reservoirs.

When quantifying emissions of carbon dioxide into the atmosphere, however, the convention is to count the mass of both the carbon atom *and* the oxygen atoms. A carbon atom has an

¹ From <https://ourworldindata.org/per-capita-co2>, retrieved October 16, 2020.

Aside 8.1: (cont.)

atomic weight of 12, while a molecule of carbon dioxide has a molecular weight of 44. Thus, 1 Gt of carbon (GtC) is equal to $44/12 = 3.67$ Gt of CO_2 (Gt CO_2) (remembering that a Gt is a gigatonne, which is a billion tonnes). The difference in these values is the mass of the oxygen atoms.

This means that emissions of carbon dioxide to the atmosphere from human activities, expressed in Chapter 5 as 11 GtC per year, would be expressed in this chapter as 40 Gt CO_2 per year. In the rest of this chapter, we'll use Gt CO_2 as our standard unit for emissions in order to be consistent with the convention of the scientific community.

We need a third factor to convert a level of total consumption, expressed in dollars, to greenhouse-gas emissions. This last factor relates how much greenhouse gas is emitted for every dollar of consumption; it is known as the *greenhouse-gas intensity*. Putting these all together, we can now relate emissions to the factors that control it in a simple equation:

$$I = P \times A \times T \quad (8.1)$$

Here I represents the total emissions of greenhouse gases into the atmosphere (these emissions then cause climate impacts, which is why emissions are represented by the letter I); P is the population, A is the affluence, and T is the greenhouse-gas intensity. The decomposition of emissions into these factors is often referred to as the *IPAT relation* or the *Kaya Identity*.

The greenhouse-gas intensity term T can be usefully broken down as the product of two terms:

$$T = EI \times CI \quad (8.2)$$

EI stands for *energy intensity* – the number of joules of energy it takes to generate one dollar of goods and services. The EI of an economy is primarily determined by two factors. First is the mix of economic activities that make up the economy. For example, it takes much more energy for a steel mill to produce one dollar's worth of steel than for a university to produce one dollar's worth of teaching. The steel mill must run blast furnaces and other heavy equipment, whereas the university only requires lighting, air-conditioning, computers, and the like. More generally, industrial manufacturing has a higher energy intensity than white-collar service-oriented activities. The more industrial manufacturing an economy has, the higher its energy intensity.

The second factor in determining the energy intensity of an economy is the efficiency with which the economy uses energy. For any economic activity, there are usually several technologies to accomplish it. For lighting, for example, there is the old-fashioned incandescent light bulb (the kind with the filament) or the LED light bulb. As described in Chapter 3, incandescent light bulbs are dreadfully inefficient, requiring 60 W of power

to produce the same light as a LED light bulb drawing 10 W. Both light-bulb technologies can light a room, but they consume vastly different amounts of energy doing it. The trade-off is that better technology is often more expensive. As a result, it takes a certain level of wealth to make an upfront investment in energy-efficient technology that will pay for itself over time. Because of this, the efficiency with which different countries utilize energy can vary greatly.

CI in Equation 8.2 stands for *carbon intensity* – the amount of greenhouse gas emitted per joule of energy generated – which reflects the mix of technologies used to generate energy, i.e., whether the economy uses fossil fuels (coal, oil, gas) or low-emission technologies (nuclear, wind, solar, etc.). Among fossil fuels, combustion of natural gas (methane or CH₄) produces the least carbon dioxide per joule of energy generated. Thus, it has the lowest carbon intensity, which is one of the reasons it is often considered to be the “greenest” of the fossil fuels. Oil produces more carbon dioxide per joule than methane, so it has a higher carbon intensity. The most carbon-intensive fossil fuel is coal – it produces roughly twice as much carbon dioxide per joule as methane – which explains why many people who are concerned with our climate are opposed to the construction of new coal-fired power plants. Energy sources that produce no carbon dioxide, and therefore have a CI of zero, include hydroelectric, nuclear, wind, and solar energy sources.

For a country such as France, which generates most of its electricity from nuclear energy, the carbon intensity will be smaller than for a country such as China or the United States, which both rely heavily on fossil fuels for electricity.

Aside 8.2: Check the units!

One of the most powerful ways to check your work is to make sure that the units in a problem work out. We do this now to close the loop on our understanding of the factors that regulate carbon emissions.

Population is obviously the number of people. Affluence is dollars of GDP per person. The product of population and affluence is therefore GDP, which has units of dollars:

$$\text{Number of people} \times \frac{\$ \text{GDP}}{\text{person}} = \$ \text{GDP}$$

Energy intensity has units of joules per dollar, and carbon intensity has units of carbon dioxide emitted per joule. Greenhouse-gas intensity is the product of energy intensity and carbon intensity, and therefore it has units of carbon dioxide emitted per dollar:

$$\frac{\text{J}}{\$ \text{GDP}} \times \frac{\text{CO}_2}{\text{J}} = \frac{\text{CO}_2}{\$ \text{GDP}}$$

Finally, the product of population, affluence, and technology has units of carbon dioxide emitted:

$$\text{Number of people} \times \frac{\$ \text{GDP}}{\text{person}} \times \frac{\text{CO}_2}{\$ \text{GDP}} = \text{CO}_2$$

8.2 How These Factors Have Changed in the Past

In the last section, emissions of carbon dioxide were deconstructed into the controlling terms: population, affluence, energy intensity, and carbon intensity. Let us look at how these terms have changed over the past few decades and how they might change in the future.

Population: Population has been rapidly increasing for most of human history. Global population reached 1 billion in 1804. The 2-billion-people mark was reached 123 years later, in 1927, and the 3-billion-people mark was reached 33 years later, in 1960. Since then, world population has been increasing by 1 billion people every 12 to 13 years, reaching 6 billion in 1999 and 7 billion in 2011. In mid-2020, when I wrote this, the population was 7.8 billion, and Figure 8.1a shows that this is 2.5 times the value in 1960. Today, world population is increasing by roughly 200,000 people per day, a population growth rate of approximately 1 percent/year. Most of this growth is occurring in the developing world, where fertility rates remain high.

Affluence: Figure 8.1a shows that affluence, measured as GDP per person, also more than doubled since 1960. This reflects my own personal experience. My teenage kids have far more material possessions (computers, TVs, cell phones) than I did as a child, and I had far more than my parents did. My parents, in turn, were far richer than my grandparents.

Energy intensity: The first part of the technology term, the energy intensity term, has markedly decreased over the past half century as our society has developed more efficient ways to use energy (Figure 8.1b). Some of this increasing efficiency has been driven by market forces: Because energy costs money, a more energy-efficient piece of equipment or process will reduce costs, which consumers want. As a result, just about everything you buy today is more energy-efficient than the comparable version of a few decades ago.

Much of this increase in efficiency is incremental, meaning that each new generation of a particular piece of equipment uses slightly less energy than the previous version. Sometimes, however, there is a revolution in technology that greatly reduces energy consumption. A good example is the revolution in lighting technology we are now experiencing. As the world switches from incandescent bulbs to LED lighting, the amount of energy being consumed by lighting will experience a substantial one-time drop.

Changes in the mix of goods and services produced by the world's economy has also led to decreases in energy intensity. Over the past century, the fraction of the world economy based on energy-intensive heavy industry and manufacturing has declined, while the fraction based on services has increased.

Carbon intensity: Figure 8.1b shows that carbon intensity, the amount of carbon dioxide released per joule of energy generated, decreased from 1960 to about 2000 as the world shifted from dirty energy, like coal, to cleaner energy sources, like natural gas. In the 2000s, coal staged a resurgence as China built out their coal fleet to support their rapid economic growth. In the 2010s, however, cheap natural gas from the development of new drilling techniques, in particular hydraulic fracturing (more commonly known as fracking) flooded

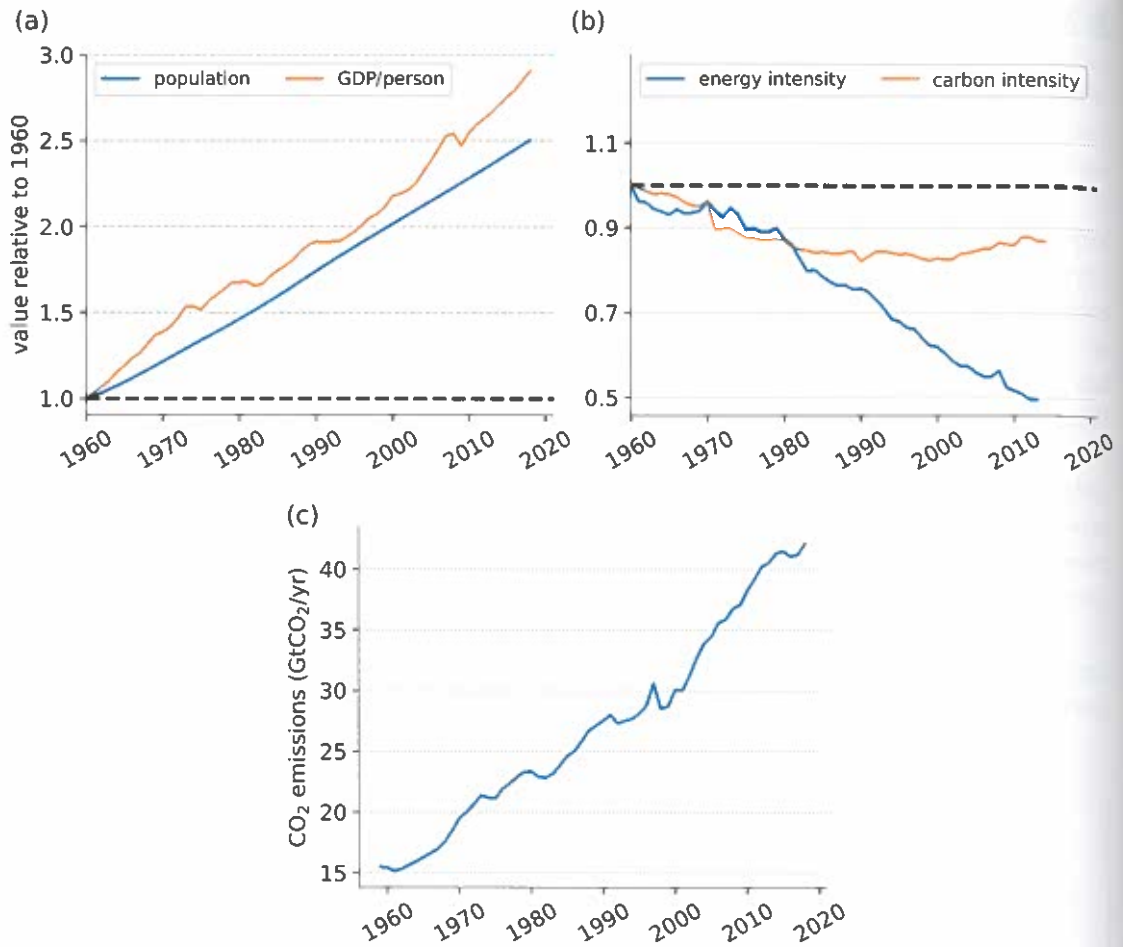


Figure 8.1 (a) Population and affluence (\$/person) values, relative to values in 1960. (b) Carbon intensity (CO_2/J) and energy intensity ($\text{J}/\text{\$}$), also relative to 1960. (c) Total emissions of carbon dioxide, in GtCO_2 per year. Data for population, GDP per person, and carbon intensity come from the World Bank, downloaded from <https://data.worldbank.org/>, retrieved June 12, 2020. Energy intensity is calculated from estimates of energy consumed and GDP downloaded from De Stercke, 2014, accessed via <http://tntcat.iiasa.ac.at/PFUDB>, retrieved June 13, 2020. Emissions data are described by Friedlingstein et al. (2019) and were downloaded from www.icos-cp.eu/global-carbon-budget-2019, retrieved June 9, 2020.

the energy market and began displacing coal, thereby reversing the trend towards increasing carbon intensity.

Combining the changes in these terms gives us the overall change in emissions. Over the past few decades, population (P) and affluence (A) have increased faster than energy intensity (EI) and carbon intensity (CI) have declined. This has led to an overall increase in emissions of nearly a factor of 3 between 1960 and 2018 (Figure 8.1c).

8.3 Emissions Scenarios

We can talk about the past all day, but the climate problem is really about the future. If our ultimate goal is to predict the future climate, we first need estimates of how much greenhouse gas human society will emit. That, in turn, requires predictions of the factors that control greenhouse-gas emissions — population, affluence, and technology.

8.3.1 Factors That Drive Emissions

But, as the old saying goes, “It’s hard to make predictions — especially about the future.”² For example, predicting future population trends requires predictions of a range of factors from the rate of poverty to the evolution of religious and social views on birth control, the rate of education of women in high-fertility regions, available healthcare in these regions, and so on.

Because of this, the community of experts does not make a single prediction of the future. Rather, they have developed a set of alternative pathways that they believe span the range of different futures the world may experience over the next century or two. These are based on five different narratives of the future, which are referred to as *Shared Socioeconomic Pathways*, usually abbreviated SSPs:³

- SSP1:** Sustainability. *The world shifts gradually, but pervasively, toward a more environmentally friendly path. The world becomes a more equal place, with economic growth in poor countries causing them to close the wealth gap with the rich world. Population growth is slow, peaking in mid-century, and the world shifts towards renewable energy.*
- SSP2:** Middle of the road. *The world follows a path in which social, economic, and technological trends are similar to historical patterns.*
- SSP3:** Regional rivalry. *The rich get richer, but the poor do not, leading to increasing conflict between regions. Nationalism is ascendant. Population growth is low in rich countries, but high in poorer ones. Consumption is resource-intensive and technological development is slow, leading to a reliance on high-carbon-intensity fuels like coal.*
- SSP4:** Inequality. *Like SSP3, the world is a divided place, but this scenario features more rapid technological development and deployment of energy technologies with low carbon intensities.*
- SSP5:** Fossil fueled development. *This world is similar to the optimistic SSP1 world, but it is powered by fossil fuels rather than SSP1’s shift towards more sustainable energy. There is an emphasis on economic growth rather than sustainability.*

For each of these narratives, scientists can make predictions of population, affluence, and technology. Population (Figure 8.2a) peaks in mid-century in all SSPs except SSP3. Affluence

² This statement has been attributed to various people, including Niels Bohr and Yogi Berra.

³ These are adapted from Riahi et al. (2017).

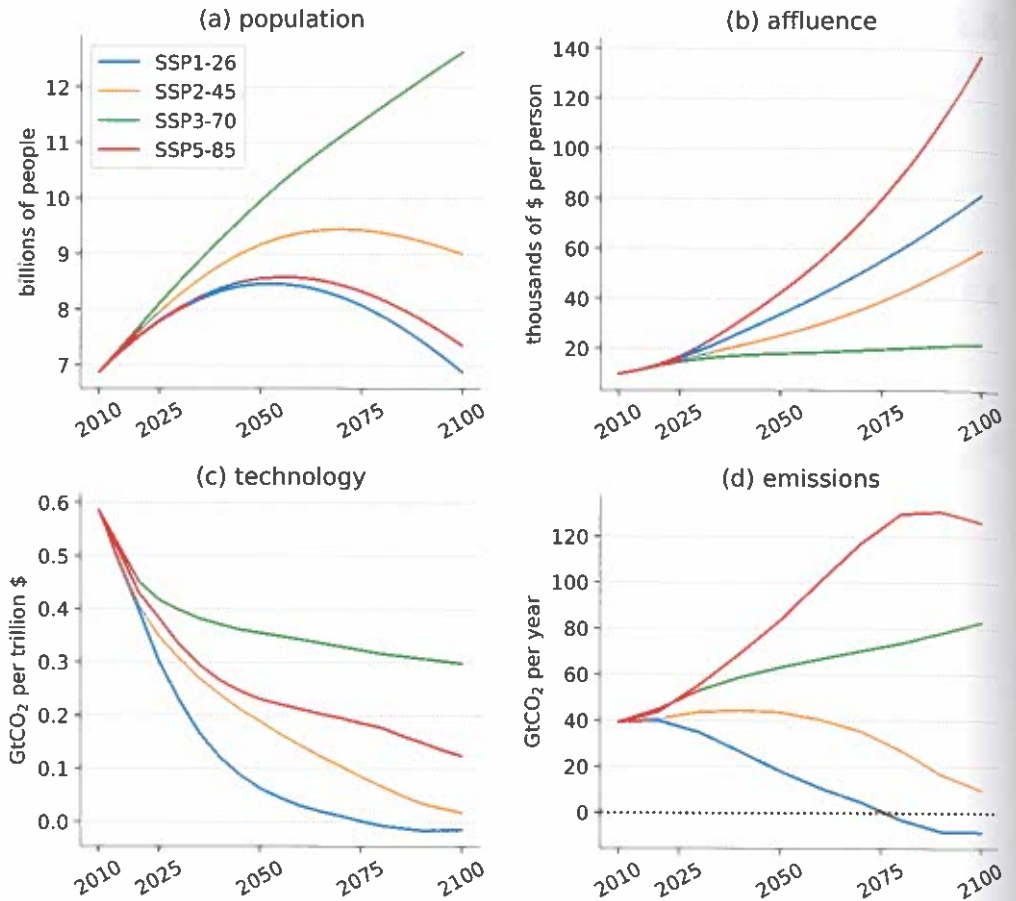


Figure 8.2 (a) Population (in billions), (b) affluence, (dollars of GDP per person), (c) technology term (GtCO_2 per trillion dollars of GDP), and (d) emissions of carbon dioxide (GtCO_2 per year). Data in panels a–c from Riahi et al. (2017), panel d from Rogelj et al. (2018) and Gidden et al. (2019), all downloaded from <https://tntcat.iiasa.ac.at/SspDb>, accessed July 4, 2020.

(Figure 8.2b) increases in all pathways, although there is considerable spread between them. SSP3 has the lowest affluence growth, about 1 percent per year, while SSP5 has the highest growth, about 3 percent per year. SSP1 also features strong growth, but it is lower than SSP5 because the SSP1 world is more focused on sustainability than consumption.

A key feature of these narratives is that they are internally consistent. For example, population growth is connected to affluence, with poor countries having higher fertility rates than rich ones. In extremely poor societies, children can be put to work at a young age and are therefore a source of income. This is generally not the case in rich countries, where children are a net drain on family resources for many years (trust me on that). In addition, high rates of childhood death in poor countries mean that parents must have many children to ensure that some of them survive into adulthood, while in richer countries with better

health care, parents can reasonably expect their children to survive into adulthood. Women tend to get better educated in richer countries, which also leads to declining fertility rates as good-paying jobs become available to women as an alternative to child rearing. Because of this, the scenarios in which the world's poor become richer (SSP1 and SSP5) feature slower population growth than the scenarios in which poverty is maintained (SSP3).

The last thing needed to obtain emissions of greenhouse gases is to determine values for the technology term, which mainly reflect the technology used to produce energy (Figure 8.2c). The units for this term are emissions per dollar, with lower values reflecting more climate-safe energy being used. In all scenarios, this term declines over time as the world relies more and more on climate-safe energy. But the rate of adoption varies significantly between the scenarios. Because the development and deployment of new climate-safe technology take money, such development and deployment are slowest in the low-affluence SSP3 world and fastest in the much richer SSP1 world. But wealth is not the only determinant. The SSP5 world is also a rich, high-growth world, but it is one that nevertheless relies predominantly on fossil fuels, so it has a larger value for this term than the SSP1 world.

8.3.2 Emissions

Given estimates of population, affluence, and technology, we can estimate emissions (Figure 8.2d). The highest emissions scenario is the SSP5, which features both high growth and a reliance on fossil fuels. In that world, 2100 emissions are 126 GtCO₂ per year, about 3 times today's emissions. The SSP3 world is actually a dirtier world, more reliant on fossil fuels. But that world is poorer than SSP5, so total emissions end up being lower. Despite that, SSP3 emissions in 2100 are more than double emissions in the late 2010s.

SSP3 and SSP5 are worlds where it is assumed that no efforts to address climate change are made beyond what the world has already done. SSP1 and SSP2 represent worlds where more strenuous efforts are made to reduce emissions. In SSP2, emissions peak around 2050 and decline thereafter. In SSP1, emissions are peaking around today and decline throughout the rest of the century. In fact, SSP1's emissions go negative around 2075, meaning that humans are pulling more carbon out of the atmosphere than they are releasing. We will discuss these negative emissions and what they mean for climate policy later in the book.

8.3.3 Atmospheric Abundance and Radiative Forcing

Given an emissions scenario, the atmospheric concentrations of carbon dioxide can be calculated by feeding those emissions into a carbon-cycle model. The carbon-cycle model calculates how much atmospheric carbon dioxide is removed by the carbon cycle and how much remains in the atmosphere (this was discussed in detail in Chapter 5).

Figure 8.3a shows atmospheric carbon dioxide abundances over the twenty-first century. As can clearly be seen, the different scenarios correspond to very different atmospheric abundances of carbon dioxide. Given these atmospheric abundances,

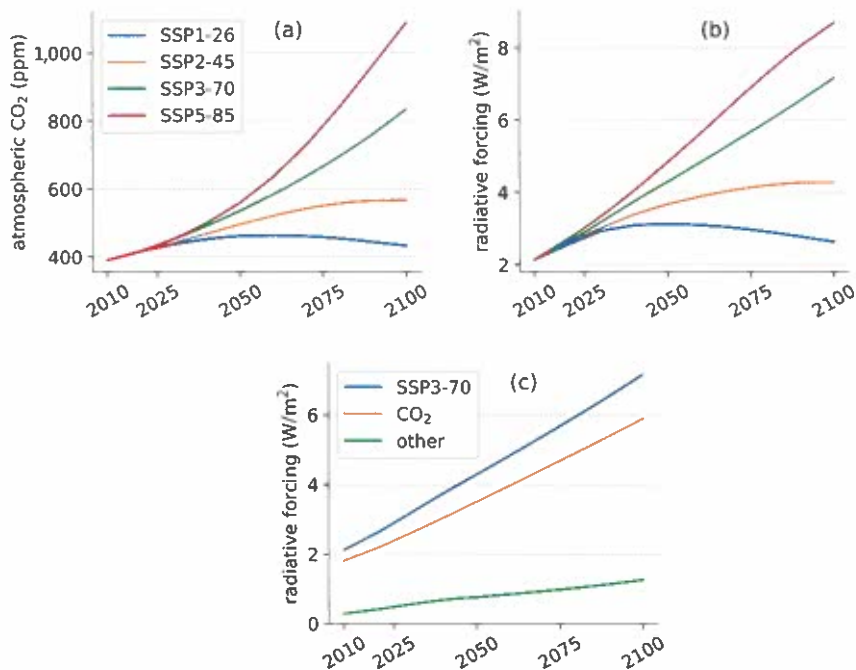


Figure 8.3 (a) Atmospheric carbon dioxide (ppm) and (b) radiative forcing (W/m^2) for four key emissions scenarios, (c) radiative forcing (W/m^2) in the SSP3–70 scenario from all constituents, from just carbon dioxide, and from everything else. This figure is based on the SSP data of Riahi et al. (2017), downloaded from the IIASA Energy Program at <https://tntcat.iiasa.ac.at/SpDb>, accessed July 4, 2020.

along with abundances of other greenhouse gases and aerosols, the radiative forcing can then be calculated. Figure 8.3b shows the radiative forcing predicted for each scenario over this century.

Aside 8.3: How are SSPs named?

Up above, I laid out the storylines for five SSPs, numbered SSP1 through SSP5. But in the figures, the pathways have two-digit numbers appended to them (e.g., SSP1–26). What does the “26” mean? It turns out that each SSP storyline is associated with a large number of possible futures, each with its own emissions profile and associated radiative forcing. There is a baseline scenario, meaning that it contains no climate policy (other than policies already adopted) as well as a number of other pathways that assume different policies to reduce emissions.

Thus, just specifying SSP1, for example, does not tell you which of the SSP1 emissions pathways I’m referring to. This problem is solved by adding the radiative forcing in 2100 to the SSP name. Thus, SSP1–26 is the emissions scenario that follows the SSP1 storyline and that has radiative forcing of 2.6 W/m^2 in 2100. This is just one of a number of SSP1 scenarios having radiative forcing in 2100 ranging from 1.9 W/m^2 to 5.0 W/m^2 .

Aside 8.3: (cont.)

In this book, I refer to the most commonly used emissions scenario from each pathway: SSP1–26, SSP2–45, SSP3–70, and SSP5–85, which have radiative forcings in 2100 of 2.6, 4.5, 7.0, and 8.5 W/m², respectively. The SSP4 pathway is less frequently analyzed, and I won't analyze it further.

Emissions scenarios are updated every decade or so as our knowledge of how society may evolve over the coming century changes. Prior to the SSPs, the set of scenarios used were known as the Representative Concentration Pathways, abbreviated RCPs. The RCPs were estimates of atmospheric concentrations of greenhouse gases over time designed to give particular trajectories of radiative forcing. The key difference from the SSPs is that there are no economic and social drivers associated with the RCPs — they are simply concentration pathways that give particular values of radiative forcing.

There were four main RCP scenarios having radiative forcing values in 2100 of 2.6, 4.5, 6.0, and 8.5 W/m² (known as RCP2.6, RCP4.5, RCP6.0, and RCP8.5). The choice of the main SSPs mirrors these values, so the RCP2.6 scenario is comparable to the SSP1–26 scenario, and the RCP4.5 and RCP8.5 scenarios are comparable to the SSP2–45 and SSP5–85 scenarios.

Figure 8.3c shows radiative forcing from the SSP3–70 scenario, along with the radiative forcing broken down into the contribution from carbon dioxide and from everything else. The plot shows that, as we go into the future, carbon dioxide is responsible for most of the increase in radiative forcing. The reason for this is carbon dioxide's long lifetime in the atmosphere (discussed in Chapter 5) — once emitted, carbon dioxide stays in the atmosphere for centuries. So carbon dioxide accumulates in the atmosphere like water in a stopped-up sink, and the radiative forcing from it accumulates too. This explains why there is such a strong focus on carbon dioxide in policy debates over climate change.

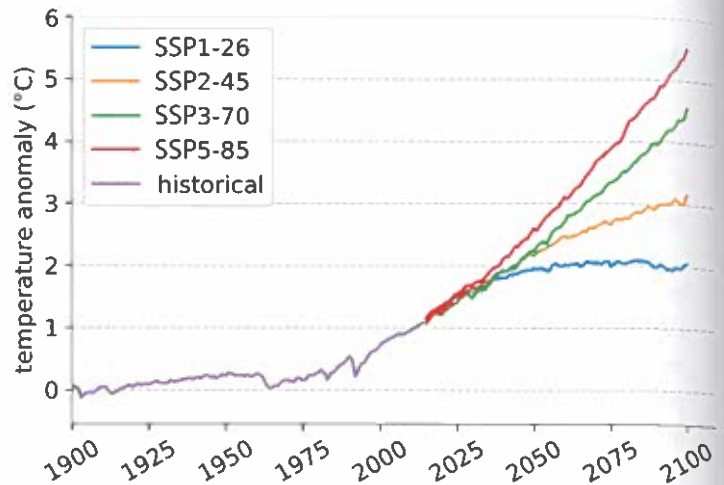
8.4 Predictions of Future Climate

8.4.1 Over the Next Century

The estimates of atmospheric radiative forcing shown in Figure 8.3 can then be translated into projected climate change for each scenario. These are plotted in Figure 8.4, and they show that the set of emissions trajectories translates into a wide range of potential future climates. The large emissions associated with the SSP5–85 scenario lead to temperature increases of 5.5°C above the pre-industrial climate in 2100 (about 4.5°C above 2020 temperature), while the low emissions associated with the SSP1–26 scenario lead to temperature increases of 2°C above pre-industrial temperatures (about 1°C above 2020 temperature).

It is worth noting that, despite huge differences in emissions (Figure 8.2d), the scenarios predict relatively similar warming until they begin to diverge around 2040–2050. This occurs because the lifetime of carbon dioxide is so long that carbon dioxide is essentially just

Figure 8.4 Average of 17 computer simulations of global and annual average surface temperature anomaly under four emissions scenarios. Temperatures prior to 2014 are from model runs driven by historical forcing. Estimates are from the CMIP6 model archive, processed by N. Swart, downloaded from <https://github.com/swartn/cmip6-gmst-anoms>, on October 16, 2020.



accumulating in the atmosphere. The amount presently in the atmosphere is large enough compared to annual emissions that it takes several decades for differences in the scenarios to translate into significant differences in the amount of carbon dioxide in the atmosphere. In 2045, for example, there is a factor of 3 difference between the highest and lowest emissions scenario, but atmospheric carbon dioxide differs by only 16 percent. The upshot is that the temperature trajectory over the next few decades has already been largely determined by emissions that have already occurred.

But Figure 8.4 also clearly shows that we do have significant control over the amount of warming experienced during the second half of the twenty-first century – there is nearly 4°C difference in temperature between the lowest (SSP1–26) and highest scenario (SSP5–85). Many of the people reading this book will live through the second half of the century, so you can look at Figure 8.4 as the range of possible futures you might experience. As we’ll talk about in the next chapter, the differences in the impacts of climate change between these possible futures are *huge*.

Aside 8.4: Will the evolution of the climate over the twenty-first century look like the trajectories plotted in Figure 8.4?

If you look at Figure 2.2, you will see that, over the past 130 years, the climate has generally warmed, but it also shows significant year-to-year variability, which is caused by things such as El Niño cycles and volcanic eruptions. Such variability means that temperatures can decline for a few years, even as the climate is experiencing a long-term warming (this was shown in Figure 2.5).

This short-term variability is missing from the projections of the future in Figure 8.4. The reason is that what is plotted in that figure is an average of many computer simulations. Individual computer simulations also show year-to-year variability that is similar to that in the

Aside 8.4: (cont.)

observations. But in the individual simulations going into the average, the highs and lows caused by the short-term variability do not occur at the same time, so when you average many simulations together, as I have done in this figure, the short-term ups and downs tend to cancel out and you get a smooth increase in temperature throughout the century. In reality, short-term variability is going to be important, and we can expect the same kinds of ups and downs seen in the past 130 years to continue to occur in the future.

8.4.2 Climate Change Beyond 2100

Even though Figure 8.4 stops in year 2100, climate change does not stop at that date. In Chapter 5, we talked about the long lifetime of carbon dioxide in our atmosphere: Of carbon dioxide emitted into the atmosphere today, about 25 percent will still be in the atmosphere in several centuries (Figure 5.8), and it will take a hundred thousand years or so to remove all of the added carbon from the atmosphere. The impact of the long residence time of carbon dioxide in our atmosphere can be seen in Figure 8.5a, which shows atmospheric carbon dioxide over the next 1,000 years for emissions scenarios in which atmospheric carbon dioxide increases until it reaches 550, 850, and 1,200 ppm, at which point emissions decline instantly to zero.

After emissions cease, carbon dioxide declines slowly. Even by the year 3000, eight to nine centuries after carbon dioxide emissions ceased, atmospheric carbon dioxide in all scenarios remains well above pre-industrial values (280 ppm). This is simply a reflection of

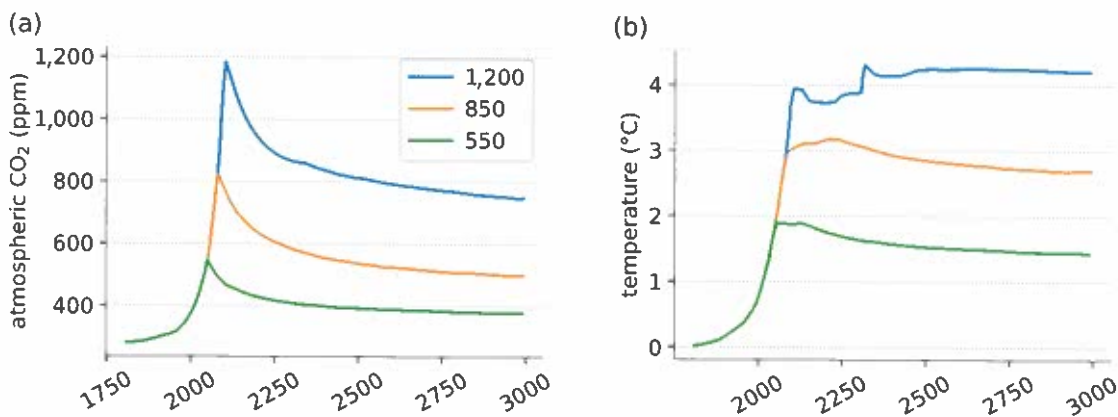


Figure 8.5 (a) Amount of carbon dioxide in the atmosphere as a function of time, for the next 1,000 years. Carbon dioxide emissions rise at 2 percent/year until it hits a peak abundance (550, 850, and 1,200 ppm); then emissions are decreased instantly to zero. (b) The temperature time series corresponding to each carbon dioxide time series. Adapted from Figure 1 of Solomon et al. (2009).

how long it takes for the carbon dioxide we're adding to the atmosphere to be removed by the carbon cycle.

The long-term evolution of temperatures associated with these carbon dioxide time series is shown in Figure 8.5b. Much like carbon dioxide, the temperatures do not significantly decline over the next 1,000 years. This is a consequence of three factors. First, carbon dioxide remains elevated throughout the millennium, so it continues to trap heat for a very long time after emissions stop. Second, the ocean's large heat capacity means that the planet cools off very slowly. This is the flip side of the situation in which the warming lags the carbon dioxide increase – the cooling will lag any decrease in atmospheric carbon dioxide abundance. Third, slow feedbacks, such as the very slow destruction of the planet's big ice sheets, will also act to oppose any cooling.

The important point here is that emitting large amounts of carbon dioxide to the atmosphere this century commits the planet to elevated temperatures for thousands of years. Once the temperatures rise, reducing emissions will not bring the temperature back down quickly. We can therefore think of climate change as being irreversible over any time period that we conceivably care about. This also means that actions we take (or do not take) to curb emissions over the next few decades will essentially determine the climate for thousands of years. It is sobering indeed to realize that people of the year 3000, 4000, or 10 000 AD will be so affected by actions we take today.

8.5 Is the Climate Predictable?

One criticism of climate predictions goes something like this: "We cannot predict the weather next week, so why does anyone believe predictions of the climate in 100 years?" This may sound reasonable, because it is based on the correct observation that weather predictions are only accurate a week or so into the future. However, the argument is built on a fatal flaw – it makes the mistake of equating weather predictions with climate predictions. In fact, it is possible to predict the climate in 100 years even if weather is only predictable for a few days.

The root cause of this conundrum is that predicting the weather and predicting the climate are fundamentally different problems. A weather forecast is a prediction of the exact state of the atmosphere at an exact time: "At 8 AM tomorrow, the temperature in Washington, DC, will be 3°C, and it will be raining." If you get the time of an event wrong – for example, you predict rain for 8 AM but it does not rain until 6 PM – then you have blown the forecast. If you predict rain for the Washington, DC, area but the rain falls 50 km to the west in Northern Virginia, then you have blown the forecast. And if your temperature is off by a few degrees and snow falls instead of rain, and it completely snarls traffic on Interstate 495, then you have *really* blown the forecast.

A climate prediction, in contrast, does not require predicting the exact state of the atmosphere at any particular time; instead, it requires predicting the *statistics* of the weather over time periods of years. Thus, a climate prediction for the month of March for the years between 2080 and 2090 for a particular location might be as follows: average monthly

temperature of 12°C , with an average high of 16°C and an average low of 5°C ; monthly average precipitation of 6.0 cm; and so on.

Being unable to predict the exact state of a complex system (e.g., the weather) does not preclude the ability to predict the statistics of the system (e.g., the climate). As an analogy, consider that it is virtually impossible to predict the outcome of a single flip of a coin. However, the statistics of coin flips are well known: If you flip a coin 10,000 times, I can tell you that you will get between 4,918 and 5,082 heads (90% confidence interval). In other words, the inability to accurately predict any single coin flip does not preclude the ability to accurately predict the long-term statistics of the coin.

To make this point more concretely, answer the following question: “Is it going to be hotter in Texas next January or next August?” If you know Texas weather, you can predict with 100 percent certainty that August is the hotter month, and you can make this prediction months, years, or decades in advance. Think about that for a minute: You just made a climate prediction that is valid years in advance – far beyond the ability to predict weather.

More technically, weather forecasts belong to a class of problems known as initial value problems. This means that, to make a good prediction of the future state of the system, you must know the state of the system now. If you have a marble rolling down a slope, and you want to predict where it will be in one second, you need to know where it is now to make that prediction. Similarly, to make a good weather forecast for tomorrow, you have to accurately know the state of the atmosphere today. The state of today’s atmosphere is then input into a forecast model, which turns out a prediction of tomorrow’s atmosphere. However, small errors in our knowledge of today’s atmosphere grow exponentially, so that a forecast more than a week or so in the future is dominated by the errors in our knowledge of today’s atmosphere. That is what sets the range of weather forecasts.

Climate forecasts are a class of problems known as boundary value problems. This type of problem does not require knowledge of today’s atmospheric state but rather requires a knowledge of the radiative forcing of the climate. This is why, for example, we can predict with 100 percent certainty that August in Texas will be on average hotter than January in Texas. We know this because we know that more sunlight falls on Texas and the rest of the northern hemisphere during summer, leading to higher temperatures.

Increases in greenhouse gases also increase the heating of the surface, although by infrared radiation rather than visible. Thus, we can have confidence that, if we add greenhouse gas to the atmosphere, the increase in surface heating will warm the planet – just as we can predict that summer will be hotter than winter.

One should not take this to mean that predicting the climate is an easier problem than predicting the weather, only that they are different problems. Some aspects of the climate problem are, in fact, harder than the weather problem. For example, because weather forecasts cover only a few days, weather models can assume that the world’s oceans and ice fields do not change. Climate models, however, cannot make this assumption, because both the world’s oceans and its ice fields can significantly change over a century. Climate models must therefore predict changes in these and other factors in order to accurately predict the evolution of the climate system over a century.

8.6 Chapter Summary

- Projections of future climate require projections of future emissions of greenhouse gases from human activities. Such projections are known as emissions scenarios.
- The factors that control emissions are population (P), affluence (A), and greenhouse-gas intensity (T). This is expressed by what is known as the IPAT relation: $I = P \times A \times T$, where I is carbon dioxide emissions. Affluence has units of dollars of economic output per person, and greenhouse-gas intensity has units of carbon dioxide emitted per dollar of economic output.
- Greenhouse-gas intensity is the product of energy intensity and carbon intensity. Energy intensity reflects the efficiency with which the society uses energy as well as the mix of economic activities in the society, with units of joules of energy consumed per dollar of economic output. The carbon intensity reflects the technologies the society uses to generate energy, and it has units of carbon dioxide emitted per joule of energy produced.
- Because predictions of the future are so uncertain, scientists have constructed a set of plausible, alternative scenarios of how the world might evolve. Taken as a group, these Shared Socioeconomic Pathways (referred to as SSPs) span the likely range of the future evolution of the world's society. Emissions scenarios have been estimated for each pathway.
- Putting these emissions scenarios into a climate model yields predictions of warming in 2100 of 2.0°C to 5.5°C above pre-industrial temperatures (1°C to 4.5°C above temperatures in the 2010s). Except in the most optimistic scenario, future warming will be much larger than the warming of ~1°C that the Earth has experienced prior to 2020.
- Climate change does not stop in year 2100. Carbon dioxide stays in the atmosphere for centuries after it is emitted, so large emissions of carbon dioxide this century will cause the Earth's temperatures to remain elevated for thousands of years. This means that the decisions we make in the *next few decades* will determine the climate for the next few millennia.
- Even though weather is not predictable beyond a few days, we can nevertheless make climate predictions decades in advance. A climate prediction is a prediction of the statistics of the system. For many complex systems, predicting the statistics is possible even in situations where predicting the specific state of the system is not.

See www.andrewdessler.com/chapter8 for additional resources for this chapter.

TERMS

Carbon intensity

Emissions scenario

Energy intensity

Greenhouse-gas intensity

Gross domestic product (GDP)
 IPAT relation
 Shared socioeconomic pathways

PROBLEMS

1. (a) Someone asks you about how much the climate will warm over the next 100 years. How do you answer?
 (b) What determines whether we are at the bottom end of the range or the top end of the predicted range?
2. (a) Define each term in the IPAT identity.
 (b) What are the units of each term? Show how the units cancel so that the I term has units of emissions of greenhouse gases.
3. (a) The T term can be broken into two terms. What are these two terms, and what are their units?
 (b) If we switch from fossil fuels to solar energy, which of the terms changes, and does this term increase or decrease?
 (c) If we convert from traditional incandescent lighting to LED lights, which of the terms changes, and does this term increase or decrease?
 (d) If we switch from natural gas to coal, which of the terms changes, and does this term increase or decrease?
4. Consider this argument: "We cannot predict the weather in a week, so there is no way we can believe a climate forecast in 100 years." Is this argument right or wrong? Explain your answer.
5. If we emit significant amounts of carbon dioxide this century, how long will the planet remain warm?
6. Assume population grows at 2 percent per year and affluence grows at 3 percent per year.
 (a) How fast does the technology term have to decrease (in percent per year) so that total emissions do not change?
 (b) How fast does the technology term have to decrease (in percent per year) to reduce emissions by 20 percent in 20 years?
7. Explain how your level of wealth impacts how much emission of carbon dioxide you are responsible for. Where do you think you fall in the spectrum of people on the Earth today? Are you a high-emissions person or a low-emissions person? What could you do to reduce your emissions?
8. In 2002, the Bush Administration set a goal of reducing greenhouse-gas intensity by 18 percent by year 2012. Estimate the historical rate of decline of greenhouse-gas intensity between 1960 and 2000 and determine whether the Bush goal was ambitious or not. Did it represent a real effort to address climate change?